

RC Autonomous Car: Budget and Options

FEB Autonomous Team

Formula Electric at Berkeley

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The ask

One 1:10-scale autonomous car on the RoboRacer (F1TENTH) standard: **\$2,300 in parts, \$2,750 with sales tax, shipping and duty**: a complete car with the RPLidar S2 and a RealSense camera bought new; \$1,951 in parts (\$2,366 all in) without the S2; \$3,199 in parts for the official reference build. Every item is listed with its shop link and today's price in Section 3. The FSAE car is on a two-year cycle and cannot host autonomous work this season. The RC car is where the team develops, tests and learns in the meantime, and the software pipeline built on it is the one that goes onto the FSAE car next year.

1 What the car is and how it drives

Part	Role
Chassis	Traxxas Fiesta ST Rally VXL, ready to run: Slash 4x4 chassis, 4WD, brushless motor, steering servo. The chassis the current RoboRacer build guide and deck drawings are made for, and the car our simulator models.
Motor controller	VESC-type. Drives motor and servo from USB and reports wheel rotation back as speed.
Computer	Runs ROS 2: sensor drivers, perception, planning, control.
2D lidar	A range per angle, 5 to 40 times a second. Sees walls and cones; enough to drive with no map.
Camera	Colour: which cone is which. Needed for the perception work.
Joystick	Bluetooth gamepad for manual driving and the dead-man switch: the car drives autonomously only while a button is held.
Power	One 3S lithium pack drives the motor; the computer takes pack voltage through a fuse; a 5 V converter feeds lidar and camera.
Deck	Laser-cut plates and standoffs from the published drawings.

Flow: lidar, wheel odometry and camera in; perception locates walls or cones; a planner picks a bearing and speed; a controller turns them into steering and throttle; the VESC actuates, gated by the dead-man. Same nodes and topic names as the team's simulator, so a stack developed there drops onto the car with only the driver nodes swapped.

Budget

Item	Now	With S2	Reference
Chassis, batteries, charger, bag	439	439	539
Computer: Jetson Orin Nano Super, NVMe, display plug and cable, antennas, USB	542	542	542
Motor controller: VESC 6 MkVI, cables, adapters	277	277	277
Power: RoboRacer power board, barrel lead, balance extender	164	164	164
Joystick	30	30	30
Lidar: RPLidar A1M8 (owned) / RPLidar S2 / Hokuyo UST-10LX	0	349	1,200
Camera: RealSense D435IF from the RealSense store	354	354	334
Screws, standoffs, zip ties	85	85	85
Spares allowance	60	60	60
Parts	1,951	2,300	3,199
Sales tax, 10.25% on the US-shipped items	160	195	287
Shipping and duty: VESC from the UK 75, power board from India 70, camera 65, domestic post 45	255	255	255
All in	2,366	2,750	3,742

Prices are what the shops showed on 24 September 2026 (Section 3). **Now** is the official build with the lidar we own: a complete car that drives laps at 1 to 2 m/s. **With S2** adds the time-of-flight RPLidar, which bolts on later without any other change; it is the one upgrade that matters. **Reference** is the official RoboRacer bill of materials at today's prices (NVIDIA raised Jetson prices by 60 to 100% in July 2026, so the computer is \$399, not \$249). The team had a RealSense last year and no longer has it, so the camera is bought new from the RealSense store, the D435IF at \$354 (its checkout total of \$455 is that price plus tax and \$65 of shipping, which sit in the two lines below the parts). Every line is the shop's list price, with tax and shipping shown separately: Alameda County sales tax at 10.25% on the US-shipped items, and a shipping and duty line for the two parts that come from abroad (the VESC from the UK: post, the 10% tariff and the carrier's brokerage fee, about \$75; the power board from India, about \$70), the camera's shipping, and domestic post from McMaster, Adafruit and NVIDIA. The S2 is bought from a US seller on Amazon, so it carries tax only. The Fiesta VXL chassis is sold out almost everywhere this month, so the list carries the in-stock 2S Fiesta at \$330; if a VXL is found, add \$100. Laser cutting and 3D printing (deck, antenna mount, lidar plate) are not in the budget: they go through the student shop account.

Exact bill of materials

Every line links to the shop page and shows the price it displayed on 24 September 2026; Amazon was checked line by line and used wherever it was cheaper and in stock. The list follows the [official RoboRacer master BOM](#) (May 2026) and the [build guide](#) step by step, with two substitutions: our RPLidar for the Hokuyo, and the in-stock 2S Fiesta for the sold-out VXL. The motor controller is the reference VESC 6 and the camera the reference RealSense in its IR-filtered variant. Not in the list, on purpose: the laser-cut deck, the printed antenna mount and lidar plate (student shop account), hex keys (in the Traxxas box), and a monitor and keyboard for first boot (borrowed).

Item	Shop	USD
Chassis and batteries		
Traxxas Ford Fiesta ST Rally 4x4 BL-2s, 74154-4 (in stock; same chassis and body as the reference; 2S-rated brushless motor, driven by our VESC at our speeds; servo and radio included)	Amazon	329.95

Item	Shop	USD
Reference part if found in stock: Traxxas Ford Fiesta ST Rally VXL, 74276-4, orange or green, 429.95 (Amazon , Crash n Bash); add 100	–	(+100)
Zeee 3S 6000 mAh 120C, EC5, 2-pack	Amazon	64.79
B6AC 80 W LiPo balance charger	Amazon	37.99
Zeee fireproof LiPo bag	Amazon	6.29
Computer		
NVIDIA Jetson Orin Nano Super Developer Kit (NVIDIA's own store, 399, stock varies week to week; SparkFun 399 on backorder; Amazon third-party 353 to 391; Seed 450.30). Wi-Fi card and two antennas are in the box	NVIDIA	399.00
NVMe SSD, 512 GB, M.2 2280 (any PCIe 3 M.2 2280 drive; flash prices are high this year)	Amazon	102.97
DisplayPort dummy plug (the Jetson boots headless with it)	Amazon	6.27
DisplayPort to HDMI cable (first setup on a borrowed monitor)	Amazon	17.99
10 dBi Wi-Fi antennas with MHF4 cables, 2-pack (external antennas on the printed mount, for range)	Amazon	8.99
Short micro-USB cables, 5-pack (Jetson to controller)	Amazon	7.19
Motor controller		
VESC 6 MkVI in aluminium box, XT90 (the reference part; ships from the UK, price from the official sheet)	Trampa	249.75
JST-PH 3-pin to header cable (controller PPM port to the steering servo)	Adafruit	1.25
Header pins (three are used on the servo lead)	Amazon	7.39
Bullet adapters, 4 mm male to 3.5 mm female, 5-pack (motor leads are 3.5 mm; check the controller's bullet size on arrival)	Amazon	8.58
EC5 female to XT90 male adapters, 3-pack (battery to controller)	Amazon	9.99
Power distribution		
RoboRacer power board, 1 unit (19 V to the Jetson, 12 V and 5 V rails, switch, fused; 125 each for three or more)	Ambimat	145.00
DC barrel extension, 5.5 x 2.5 mm (power board to Jetson)	Amazon	9.99
3S balance plug extension leads, 5-pack (battery to power board)	Amazon	8.99
Manual control		
8BitDo Ultimate 2C wireless gamepad (dead-man switch and manual driving)	Amazon	29.99
Sensors		
RPLidar A1M8, owned (spec)	–	0
Intel RealSense D435IF (the official D435i with an IR-pass filter, so depth also works outdoors and on the FSAE car; in stock; checkout shows 455.00 with tax and shipping, counted in the lines below)	RealSense	354.00
Hardware		
M2 to M5 screw assortment, 4 to 20 mm (covers the M3 deck screws and the M5 controller screws)	Amazon	21.99
M3 hex standoffs, 14 mm	McMaster	3.68
M3 hex standoffs, 25 mm	McMaster	10.76
M3 hex standoffs, 45 mm (deck to chassis, antenna bracket)	McMaster	27.90
M2.5 x 6 mm screws (Jetson)	McMaster	5.30
M2.5 standoffs, 10 mm (Jetson)	McMaster	2.16
Zip ties, 4 inch (cables away from the driveshaft)	Amazon	12.99
Spares		
Allowance: bumper, servo saver, tyres, pinion, from the Traxxas parts finder as they break	Traxxas	60.00
Now, parts		1,951
Upgrades		
RPLidar S2, 30 m time of flight (Slamtec listing, in stock, ships within the US; Seed 398.99 with a US-warehouse option and RobotShop are the fallbacks)	Amazon	349.00
Now plus S2, parts		2,300
Sales tax, 10.25% on the US-shipped items (1,905 taxable; the VESC and power board are not)		195

Item	Shop	USD
Shipping and duty: VESC from the UK 75, power board from India 70, camera 65, domestic post 45		255
All in, now plus S2		2,750
All in, without the S2		2,366

Does it cover the whole build? Yes. Walking the official guide: lower chassis (remove the Traxxas ESC and receiver, keep motor and servo, three 45 mm standoffs, battery), upper deck (controller on M5 screws, antenna bracket on two 45 mm standoffs, Jetson on M2.5 standoffs, power board on 25 mm standoffs, lidar on four M3 screws), wiring (three bullet adapters on the motor leads, battery to controller through the EC5 to XT90 adapter, balance lead to the power board, barrel lead to the Jetson, servo on the PPM cable, lidar and controller on USB, zip ties away from the driveshaft), then software on the NVMe. The Orin Nano developer kit has four USB-A ports: lidar, controller, camera and joystick receiver fill them exactly; a hub is only needed for a fifth device.

Where the money could be saved, and why we do not. A [Flipsky Mini FSESC 4.20 50 A](#) (\$72, 3S to 13S, VESC 4.12 hardware, runs the same firmware) in place of the VESC 6 MkVI takes \$178 off every total: \$1,773 now, \$2,122 with the S2, in parts. It is what many student builds use, but it does not match the deck's mounting holes or the sheet's connectors, so it means drilling, adapters and a week of surprises. We keep the reference part. (The Flipsky Mini FSESC 6.7 is 14 V minimum and cannot be used at all.) The S2 lidar is best bought as a second, later request: the car does not wait for it.

What the A1M8 costs us. Same ROS message, same software, a printed plate instead of the Hokuyo bolt pattern, USB power instead of 12 V, so nothing about the build gets harder. What changes is the ceiling: 8,000 samples a second at 5 to 10 Hz gives about 1° resolution, ranges reliable to roughly 6 m indoors, and a scan that takes 100 to 180 ms while the car moves, so the laps are 1 to 2 m/s and the maps are coarse. Cones at 6 m give two to four returns, enough for the cone pipeline. Triangulation lidars also wash out in direct sunlight, so the A1M8 car stays indoors. The S2 lifts all of that (30 m, 32,000 samples a second, works outdoors) and is the one part that makes the car fast.

4 Options in each area

Lidar

The lidar sets the speed the car can drive: at 10 Hz it moves 25 cm between scans at 3 m/s. All publish the same ROS message, so a swap changes no code; the RPLidar sits on a small printed adapter plate and the Hokuyo bolts straight to the deck.

Lidar	USD	Rate	Range	Reality
RPLidar A1M8 (owned)	0	5 to 10 Hz	12 m	laps at 1 to 2 m/s; 1° resolution, coarse maps; runs the car until the S2 arrives
RPLidar A2M12	229	12 Hz	12 m	fine indoors; noisy on dark surfaces
RPLidar S2	399	10 Hz	30 m	time of flight; clean maps; to 3 m/s
RPLidar A3	599	15 to 20 Hz	25 m	fastest cheap unit
Hokuyo UST-10LX	1,200	40 Hz	10 m	the reference; what race winners run

Compute

Computer	USD	On the RC car	On the FSAE car
Raspberry Pi 5 , 8 GB	130	lidar-only driving; no camera perception	no
Jetson Orin Nano Super , 8 GB	399 + 100 NVMe	the official part; fits the deck; enough for everything the car does	no: a 3D lidar and several cameras need more
Jetson Orin NX 16 GB on the Nano carrier	about 1,200	fits; overkill	a first driverless stack: M1 lidar, two cameras, detector
Jetson AGX Xavier 16 GB (owned, 2018)	0	no: 630 g box, does not fit the deck	a bridge for one or two seasons; last software is JetPack 5
Jetson AGX Orin 64 GB	3,500	no: 1.4 kg	the full answer, when the stack needs it

Recommended: the Orin Nano on the RC car, permanently. The FSAE computer is decided next year with a season of data; the owned Xavier carries FSAE perception development until then. If the club prefers one purchase for both cars, the Orin NX is the largest that fits the RC car and moves to the FSAE car later, at about \$800 more, leaving the RC car needing a Nano then.

Why a Jetson from day one. Jetson to Jetson is the same software: same operating system image, same ARM64 build, same CUDA and TensorRT, same ROS 2 packages. Moving the stack from the Orin Nano to an Orin NX or AGX Orin is copying it across. Developing on an x86 laptop or PC instead means a different architecture, different driver and CUDA versions, and recompiling everything, with the porting landing at the worst moment, when the FSAE car is finally ready. Building on the Nano now makes the later upgrade trivial.

A separate line for the on-car FSAE computer. If the budget process allows it, the FSAE car's computer should be its own item, requested once the team has a season of data on what the stack needs: about \$1,200 for an Orin NX 16 GB on a carrier board or about \$3,500 for an AGX Orin developer kit, at the July 2026 prices. It is not part of this ask, and buying it now would mean guessing. Two-year total, RC car with the S2 now plus the FSAE computer next year: about \$3,150 with an Orin NX, about \$5,450 with an AGX Orin.

Camera

The cone pipeline needs colour; range comes from the lidar. The [RealSense D435i](#), is the official part and the FSAE car's camera, so the same driver and calibration carry over. The team had one last year and no longer does, so the budget buys the D435IF variant: the same camera with an infrared-pass filter, which keeps depth usable in sunlight for the FSAE car; \$354 at the RealSense store, \$455 with tax and shipping. A [Logitech C920](#), \$67, would do the RC car's cone job alone if the camera had to be cut. [OAK-D Lite](#), \$269, stereo depth and on-camera inference, is the third option. [OAK-D Pro W](#) or [ZED 2i](#), \$400 to \$500, global shutter and weatherproof: FSAE-grade cameras that also fit the RC car, if the club wants the camera bought once.

Controller

The controller must accept the car's 3S pack (9 to 12.6 V). The [VESC 6 MkVI](#), \$250, is the reference part: it fits the deck, matches the sheet's cables and adapters, and is what the build guide photographs. The [Flipsky Mini FSESC 4.20](#), \$72, runs the same firmware on older hardware and would save \$178 at the cost of drilling the deck and sorting connectors; noted, not chosen. The popular Flipsky Mini FSESC 6.7 is rated 14 V minimum (4S) and does *not* suit this car.

5 What we already own, and where it goes

- **RPLidar A1M8.** Onto the RC car: laps at 1 to 2 m/s until the S2 arrives, then a spare. USB-powered, four screws on a printed plate.
- **RoboSense M1 3D lidar.** It can be bolted to the RC car on a custom plate, and the car would drive, badly: 0.7 kg on a 3 kg car spoils the handling, its half-metre blind zone hides the nearest wall in a 2 m corridor, and its 120° forward view gives wall-following nothing to work with. It is an FSAE sensor; with the Xavier it forms a perception rig for cone-track data this semester.
- **AGX Xavier 16 GB.** Computer for that rig and a bridge for the FSAE car; not for the RC car.
- **RealSense D435i.** The team had one last year; it is no longer with the team, so the budget replaces it.
- **Tamiya 1:10 chassis.** Motorless touring kit: no drawings, no match to the simulator, a semester of mechanical work to save \$300. Bench mule.
- **Alameda test track.** Outdoor sessions for the rig and, later, the FSAE car; not needed for the RC car, which runs in a corridor.

6 Why it is worth it

This semester

- The FSAE car is being built and cannot host autonomy; without a platform the team has nothing to run on until next year.
- Members stay for what they can see move.
- Real dynamics and real sensors cannot be simulated: grip, servo backlash, battery sag, a lidar that sees chair legs.
- New members learn wiring, power, safety, motor control and ROS 2 on hardware at low cost and low risk.
- The simulator and the car share one codebase, so every member's stack has somewhere to go.

Carrying over to the FSAE car

- **Same pipeline.** Cone detection with camera plus lidar, midline between cones, raceline, tracking controller: the Formula Student Driverless stack at small scale.
- **Same tools.** ROS 2 nodes and topics, sensor drivers, bag recording and replay, sim-to-real testing, safety procedures.
- **Same method.** Develop in the simulator, transfer to the car, measure the gap, close it.
- **What does not transfer:** vehicle dynamics and tuned parameters. A 3 kg car says nothing about a 250 kg car's tyres.

7 Will it work, and the practical realities

The build is the F1TENTH/RoboRacer platform, assembled by hundreds of university teams since 2016 from a public build guide; the parts list is that guide's own with the sensors we own and a cheaper controller. Driving laps is a solved problem, and our simulator already runs the same car and software. The car can run the whole pipeline: mapping, localisation, a raceline and a proper controller; the cheaper sensors cap speed, not capability. The recommended car is eligible for RoboRacer physical events as built; a lidar upgrade is what would make it fast.

- **One owner.** The car needs a named person for assembly, batteries and access, or it goes unused. This is the main risk.
- **Time.** Two to four weekends to assemble and integrate; first autonomous lap within a month; mapping, racelines and perception fill the semester.
- **Speed.** 1 to 2 m/s on the A1M8, about 3 m/s on the S2, faster only with the Hokuyo. The simulator car goes faster; members will feel the gap.
- **Access.** One car; time on it is earned through the simulator's qualification rule.
- **Batteries.** LiPo bag, supervised charging, a written procedure before the first charge.
- **Wear.** Bumpers and servo savers break; the spares line covers it.
- **Not in this budget.** A 1:5-scale car (Traxxas X-Maxx, Arrma Kraton 8S, Losi 5IVE-T: about \$3,500 even with the M1 and Alameda counted, and not runnable on campus), a second car, and the FSAE computer. All later decisions.

RoboRacer build guide: <https://f1tenth.readthedocs.io>; official master BOM: [Google Sheet](#)
Team simulator and manuals: <https://github.com/Pranman1/feb-racing>